Questions

1. (a) The vectors

$$\mathbf{p} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} \quad \text{and} \quad \mathbf{q} = \begin{pmatrix} 2 \\ m \\ -1 \end{pmatrix}$$

span a plane with normal vector

$$\mathbf{n} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

Given that $\mathbf{q} \times \mathbf{p}$ is in the same direction as $\mathbf{n}$, find m and the positive constant λ such that

$$\mathbf{q} \times \mathbf{p} = \lambda \mathbf{n}$$ **[3]**

(b) Using $\mathbf{q} \times \mathbf{p}$, determine the acute angle between $\mathbf{p}$ and $\mathbf{q}$. **[3]**

2. You are given the fixed points $A(1, 0, 1)$ and $C(4, 0, 2)$, and the variable point $B(t, 2, 5 - t)$, where t is a real parameter.

(a) Using only the geometrical properties of the vector product, explain why the statement "$\overrightarrow{AB} \times \overrightarrow{AC} = 0$" is false for all values of t. **[2]**

(b) (i) Use the vector product to find an expression, in terms of t, for the area of triangle ABC. **[4]**

(ii) Hence determine the value of t for which the area of triangle ABC is a minimum. **[2]**

3. With respect to a fixed origin O, the points A and B have coordinates $(1, 1, 0)$ and $(p, 3, 2)$ respectively, where p is a constant.

For each of the following, determine the possible values of p for which

(a) $\overrightarrow{AB}$ makes an angle of $60°$ with the positive y-axis. **[3]**

(b) $\overrightarrow{OA} \times \overrightarrow{AB}$ is parallel to

$$\begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

[3]

(c) the area of triangle OAB is $\frac{3}{2}$. **[3]**

4. Points P, Q and R have position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ respectively, relative to origin O.

It is given that $(\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q}) = 2\mathbf{p}$ and that $|\mathbf{p}| = 5$.

(a) Determine each of the following as either a single vector or a scalar quantity.

(i) $(\mathbf{r}-\mathbf{q})\times(\mathbf{q}-\mathbf{p})$ **[1]**

(ii) $\mathbf{p}\times\big((\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})\big)$ **[2]**

(iii) $\mathbf{p}\cdot\big((\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q})\big)$ **[2]**

(b) Describe a geometrical relationship between the points O, P, Q and R which can be deduced from

(i) the statement $(\mathbf{q}-\mathbf{p})\times(\mathbf{r}-\mathbf{q}) = 2\mathbf{p}$ **[1]**

(ii) the result of part (a)(iii) **[1]**

5. **(a)** Let

$$\mathbf{p} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \quad \text{and} \quad \mathbf{q} = \begin{pmatrix} x \\ 0 \\ 1 \end{pmatrix}$$

The area of the parallelogram formed by $\mathbf{p}$ and $\mathbf{q}$ is $\sqrt{2}$.
It is also given that $\mathbf{p} \times \mathbf{q}$ is perpendicular to

$$\begin{pmatrix} 1 \\ 3 \\ m \end{pmatrix}$$

Find x and m. **[4]**

(b) Hence, using $|\mathbf{p} \times \mathbf{q}|$, determine the acute angle between $\mathbf{p}$ and $\mathbf{q}$. **[3]**

6. The point P has coordinates $P(2, -1, 1)$. The vector $\overrightarrow{PQ} = (1, 2, 0)$ and the midpoint of PR is $M(1, 0, 2)$.

(a) Use a vector product to show that the area of triangle PQR is $\sqrt{14}$. **[4]**

(b) Find a vector equation of the plane containing P, Q and R in the form $\mathbf{r} \cdot \mathbf{n} = k$. **[1]**

(c) Hence find the exact distance of the plane from the origin. **[1]**

7. The points P, Q and R have position vectors $\mathbf{p} = 2\mathbf{i} - 2\mathbf{k}$, $\mathbf{q} = 3\mathbf{j} + 3\mathbf{k}$ and $\mathbf{r} = t\mathbf{i} - \mathbf{j} + \mathbf{k}$ respectively, relative to the origin O.

Determine the value(s) of t in each of the following cases.

(a) The line OR is parallel to a normal to the plane OPQ. **[2]**

(b) The volume of tetrahedron $OPQR$ is 9. **[4]**

8. The points A, B and C lie on the plane Π.

The position vectors of A and B are $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i} + \mathbf{j}$ respectively.

The centroid G of triangle ABC has position vector $4\mathbf{i} + \mathbf{j} + 2\mathbf{k}$.

(a) Find $\overrightarrow{AB} \times \overrightarrow{AC}$. **[4]**

(b) Find an equation for Π in the form $\mathbf{r} \cdot \mathbf{n} = p$. **[2]**

The point D has position vector $\mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$.

(c) Determine the volume of the tetrahedron $ABCD$. **[4]**

9. With respect to a fixed origin O the point A has coordinates $(9, 10, -1)$ and the line ℓ is the intersection of the planes $x - z = -2$ and $x + 2y = 5$.

The points B and C lie on ℓ such that $AB = AC = 15$.

Given that A does not lie on ℓ and that the z coordinate of B is negative

(a) determine the coordinates of B and the coordinates of C. **[4]**

(b) Hence determine a Cartesian equation of the plane containing the points A, B and C. **[3]**

The point D has coordinates $(8, 4, \alpha)$, where α is a constant.

Given that the volume of the tetrahedron $ABCD$ is 156

(c) determine the possible values of α. **[4]**

Given that $\alpha > 0$

(d) determine the shortest distance between the line ℓ and the line passing through the points A and D, giving your answer to 2 significant figures. **[4]**

10. The lines L_1, L_2 and L_3 are given by

$$L_1: \quad \mathbf{r} = \begin{pmatrix} 8 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$$

$$L_2: \quad x = 9 - 4\mu,\ y = -2 + 3\mu,\ z = -1 + 3\mu$$

$$L_3: \quad \frac{x-3}{1} = \frac{y+6}{-5} = \frac{z-1}{-2}$$

The pairwise intersections of these lines are the vertices of a triangle.

Find the area of this triangle. **[9]**